PROJECT REPORT

A22 – INTERCONNECTED SYSTEM FREQUENCY RESPONSE GUI

Signals and Systems | MATLAB GUI Project

Submitted by

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Abstract

This project studies the frequency responses of interconnected digital systems using MATLAB. It considers a low-pass filter (LPF), a high-pass filter (HPF), and an all-pass reference response. The project compares parallel and series connections, plots their frequency responses, and examines how cutoff frequency affects the similarity between the parallel LPF–HPF response and an ideal all-pass response. The report includes the theoretical background, analytical derivations, GUI operation, algorithm, applications, advantages, limitations, and conclusion.

1. Introduction

A digital system processes an input signal according to its system function. The frequency response explains how the magnitude and phase of a sinusoidal component change with frequency. In a parallel connection, the system outputs are added; in a series (cascade) connection, the transfer functions are multiplied. MATLAB can be used to calculate and visualize these responses.

2. Objectives

- To understand LPF, HPF, and all-pass frequency responses.
- To study parallel and series interconnections of systems.
- To plot and compare responses over the frequency range $0 \le \omega \le \pi$.
- To examine whether the parallel LPF–HPF response behaves like an ideal all-pass system.
- To find the cutoff setting that minimizes deviation from an all-pass reference for a selected filter design.
- To answer the viva question about the series response at $\omega = \pi/2$.

3. Software and System Requirements

- Software: MATLAB.
- Hardware: A computer capable of running MATLAB.
- GUI elements: Figure window, cutoff controls, push buttons, axes, and result labels.
- No external hardware is required.

4. Theoretical Background

4.1 Low-Pass Filter (LPF)

A low-pass filter passes low-frequency components and attenuates higher-frequency components. Its magnitude response is generally high near $\omega = 0$ and decreases as frequency increases.

4.2 High-Pass Filter (HPF)

A high-pass filter attenuates low-frequency components and passes higher-frequency components. Its magnitude response is generally low near $\omega = 0$ and rises toward higher frequencies.

4.3 All-Pass System

An ideal all-pass system has constant magnitude at every frequency, commonly $|H_{AP}(e^{j\omega})| = 1$. Its phase response can still vary with frequency.

4.4 Interconnected Systems

$$\text{Parallel connection: } H_{par}(z) = H_1(z) + H_2(z)$$

$$\text{Series connection: } H_{ser}(z) = H_1(z) \times H_2(z)$$

For a series connection, $|H_{ser}(e^{j\omega})| = |H_1(e^{j\omega})| \times |H_2(e^{j\omega})|$. For a parallel connection, the complex frequency responses are added; their magnitudes cannot generally be added directly.

5. System Model

For a reproducible first-order example, let $0 < a < 1$ and define:

$$H_{LPF}(z) = (1 - a) / (1 - a z^{-1})$$

$$H_{HPF}(z) = a(1 - z^{-1}) / (1 - a z^{-1})$$

For these particular complementary definitions:

$$H_{LPF}(z) + H_{HPF}(z) = 1$$

Thus, this specific pair has an exactly unity parallel response for every valid a. The general statement that an LPF and HPF do not necessarily form an all-pass response applies when the filter designs or cutoff settings are not complementary. The selected equations must therefore be stated clearly.

6. Analytical Derivation

6.1 Parallel Connection

If X(z) is the input and the two systems are connected in parallel, then:

$$Y(z) = H_1(z)X(z) + H_2(z)X(z)$$

$$H_{par}(z) = Y(z)/X(z) = H_1(z) + H_2(z)$$

For the complementary example in Section 5, $H_{par}(z) = 1$ and its magnitude is unity at every frequency.

6.2 Series Connection

In a cascade, the output of the first system becomes the input to the second system. Hence:

$$H_{ser}(z) = H_1(z)H_2(z)$$

$$|H_{ser}(e^{j\omega})| = |H_1(e^{j\omega})|\,|H_2(e^{j\omega})|$$

The product generally varies with frequency and is not necessarily an all-pass response.

6.3 Cutoff Search and Deviation

For each cutoff parameter a, evaluate the parallel response over a frequency grid and calculate its mean-square deviation from an ideal all-pass magnitude:

$$D(a) = \text{mean}_{\omega}[(|H_{par}(e^{j\omega})| - 1)^2]$$

The value of a that minimizes D(a) gives the closest match for the selected model. For the complementary equations in Section 5, D(a) is zero for every valid a, so there is no unique best cutoff. A unique optimum requires a specified non-complementary filter design or another

constraint.

7. Viva Question: Is the Series Response Known at ω = π/2?

For a series connection, the frequency response at ω = π/2 is:

$$H_ser(e^{(j\pi/2)}) = H_1(e^{(j\pi/2)}) \times H_2(e^{(j\pi/2)})$$

Since $e^{(-j\pi/2)} = -j$, substitute $z^{-1} = -j$ into each selected transfer function and multiply the resulting complex values. The numerical answer depends on the chosen filter coefficients and exact LPF/HPF equations.

8. GUI Design and Operation

The intended GUI provides controls for the cutoff parameter, a button to calculate or plot the responses, and axes for the LPF, HPF, parallel, series, and all-pass reference responses. A second plot can display the deviation from the all-pass magnitude against cutoff frequency.

- Select or enter the cutoff parameter.
- Calculate $H_1(e^{(j\omega)})$ and $H_2(e^{(j\omega)})$ over $0 \le \omega \le \pi$.
- Compute the parallel response by adding the complex responses.
- Compute the series response by multiplying the complex responses.
- Plot the individual and combined magnitude responses with the all-pass reference.
- Sweep cutoff values, calculate deviation, and display the minimum.

9. Algorithm

1. Start MATLAB and create the GUI.
2. Define a frequency vector from 0 to π.
3. Choose the filter model and cutoff parameter.
4. Evaluate the complex responses H_1 and H_2.
5. Calculate $H_par = H_1 + H_2$ and $H_ser = H_1 \times H_2$.
6. Plot the LPF, HPF, parallel, series, and all-pass responses.
7. Sweep cutoff values and calculate the deviation for each value.
8. Display the cutoff with minimum deviation and stop.

10. Applications

- Learning digital signal-processing concepts in laboratories.
- Studying filter combinations in audio and communication systems.
- Comparing frequency responses and cutoff effects visually.
- Understanding parallel and cascaded signal-processing blocks.

11. Advantages and Limitations

Advantages

- Plots make filter behaviour easier to understand.
- Parallel and series calculations can be compared in one interface.
- Cutoff sweeps help find the best fit for a chosen filter design.
- Useful for demonstrating frequency response and system interconnection.

Limitations

- Results depend on the selected filter equations and coefficients.
- A complementary LPF–HPF pair can sum to unity for every cutoff parameter, so a unique optimum may not exist.
- An all-pass system has constant magnitude, but its phase may vary.
- The model is educational and may not represent every practical filter.

12. Conclusion

The A22 Interconnected System Frequency Response GUI demonstrates LPF and HPF frequency responses and their parallel and series connections. Parallel transfer functions add, whereas series transfer functions multiply. A cutoff-deviation plot can identify the closest all-pass match for a specified filter design. The exact filter equations must be stated because complementary LPF and HPF designs may produce an exactly unity parallel response.

13. References

- Signals and Systems course notes on frequency response and system interconnections.
- MATLAB documentation on plotting, GUI controls, and frequency-response calculations.

14. MATLAB GUI Output

The following figure shows the frequency-response plots for the low-pass filter (LPF), high-pass filter (HPF), and all-pass system.

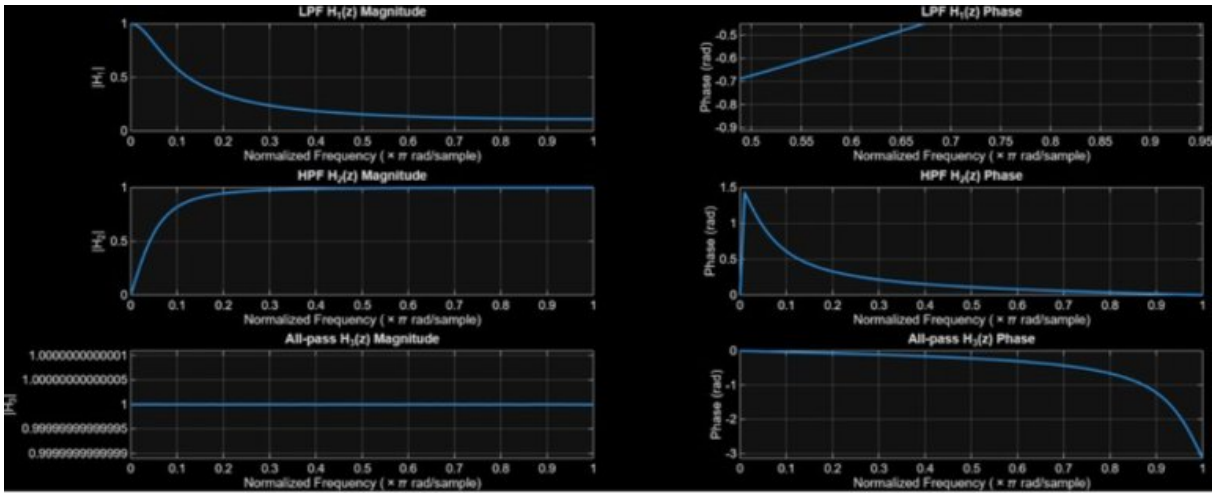


Figure 1: Magnitude and phase responses of LPF, HPF, and all-pass systems